

Lecture 14: Related Rates (Perimeter/Area/Volume Relations)

Goal: If a quantity changes over time, explain how its derivative changes over time.

1 Introduction (Circles)

Problem 1. A circle's area and radius are related by the equation:

$$A = \pi r^2$$

If A and r are changing as time advances, is there any relation between $\frac{dA}{dt}$ and $\frac{dr}{dt}$?



$$A = \pi r^2$$

$$A(t) = \pi [r(t)]^2$$

$$\frac{d}{dt}(A(t)) = \frac{d}{dt}(\pi [r(t)]^2)$$

$$\frac{dA}{dt} = \pi 2(r(t)) \frac{dr}{dt}$$

$$\frac{dA}{dt} = 2\pi r \frac{dr}{dt}$$

Problem 2. (a) In the previous example, if $r = 2$ and $\frac{dr}{dt} = 3$, then what is the value of $\frac{dA}{dt}$? Interpret.

$$\frac{dA}{dt} = 2\pi r \frac{dr}{dt}$$

$$\left. \frac{dA}{dt} \right|_{\substack{r=2 \\ r'=3}} = 2\pi(2)(3) \\ = 12\pi$$

When the radius is 2 units, the area is increasing at a rate of 12π units²/sec.

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(b) In the previous example, if $r = 1$ and $\frac{dA}{dt} = 2\pi$, then what is the value of $\frac{dr}{dt}$?



$$\begin{aligned}\frac{dA}{dt} &= 2\pi r \frac{dr}{dt} \\ 2\pi &= 2\pi (1) \frac{dr}{dt} \\ 2\pi &= 2\pi \frac{dr}{dt} \\ \frac{dr}{dt} &= 1\end{aligned}$$

If the radius is 1 unit, in order for the area to change at a rate of 2π units²/sec, the radius will need to change at a rate of 1 u/s.

Problem 3. The radius of a circle r is increasing at a constant rate 3 cm/min.

(a) Find the rate of change of the area of the circle (A) when the radius is 5cm.



$A = \text{area}$
Formula:
 $A = \pi r^2$

K:

$$\frac{dr}{dt} = 3$$

NTK:

$$\frac{dA}{dt} \text{ when } r = 5$$

$$A = \pi r^2$$

$$\frac{dA}{dt} = 2\pi r \frac{dr}{dt}$$

$$\frac{dA}{dt} = 2\pi (5)(3) = 30\pi$$

"When the radius is 5 cm, the area of the circle is increasing at a rate of 30π cm²/min."

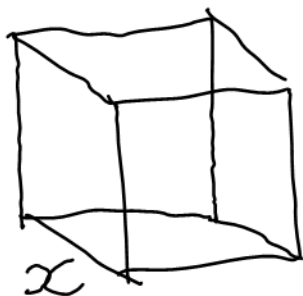
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- (b) Find the rate of change of the circumference of the circle (C) when the radius is 5cm.

2 Rectangular Prisms

Problem 4. The edges of a cube are shrinking at a rate of 10 cm/s.

- (a) How fast is the volume (V) shrinking when each side length is 9cm long?



x = side lengths
 V = Volume

Formula:

$$V = x^3$$

K:

$$\frac{dx}{dt} = -10$$

NTK:

$$\frac{dV}{dt} \text{ when } x=9$$

Here is when
you would want to
plug in
numbers

$$V = [x(t)]^3$$

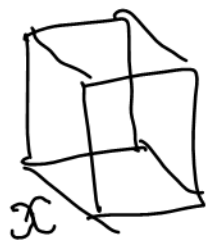
$$\frac{dV}{dt} = 3x^2 \frac{dx}{dt}$$

$$\left. \frac{dV}{dt} \right|_{x=9, x'=-10} = 3(9)^2(-10) = -2430$$

When the side lengths are 9cm, the volume is shrinking at a rate of $2430 \text{ cm}^3/\text{s}$

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(b) How fast is the surface area (A) shrinking when each side length is 9cm long?



$A =$ surface area of the cube

Formula:

$$A = 6x^2$$

K:

$$\frac{dx}{dt} = -10$$

NTK:

$$\frac{dA}{dt} \text{ when } x=9$$

$$A = 6x^2$$

$$\frac{dA}{dt} = 12x \frac{dx}{dt}$$

$$\left. \frac{dA}{dt} \right|_{\substack{x=9 \\ x'=-10}} = 12(9)(-10)$$

$$= -1080$$

When the side lengths of the cube are 9 cm long, the surface area of the cube is decreasing at a rate of 1080 cm²/s.

3 Spheres

Problem 5. A balloon is (roughly) a sphere. The balloon deflates and its radius decreases at a rate of 2 cm/s.

(a) How fast is the volume (V) shrinking when the radius is 5cm long?



$r =$ radius

$V =$ Volume

Formula:

$$V = \frac{4}{3}\pi r^3$$

K:

$$\frac{dr}{dt} = -2$$

NTK:

$$\frac{dV}{dt} \text{ when } r=5$$

$$V = \frac{4}{3}\pi r^3$$

$$\frac{dV}{dt} = \frac{4}{3}\pi (3r^2 \frac{dr}{dt})$$

$$\frac{dV}{dt} = 4\pi r^2 \frac{dr}{dt}$$

$$\left. \frac{dV}{dt} \right|_{\substack{r=5 \\ r'=-2}} = 4\pi(5)^2(-2)$$

$$= -200\pi$$

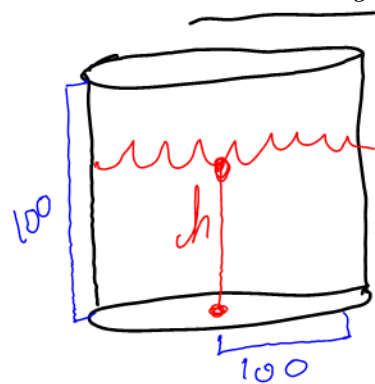
When its radius is 5 cm, the balloon's volume is decreasing at a rate of 200π cm³/s.

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(b) How fast is the surface area (A) shrinking when the radius is 5cm long?

4 Cylinders

Problem 6. A cylindrical tank with a radius and height of 100 cm stands upright. Water is being drained at a rate of $7 \text{ cm}^3/\text{s}$. How fast is the water level changing when the tank is half empty.



h = Water Level
 V = Volume / Amount of Water

Formula:

$$V = \pi r^2 h$$

$$V = 10000\pi h$$

K:

$$\frac{dV}{dt} = -7$$

NTK:

$\frac{dh}{dt}$ when

$$h = \frac{1}{2}(100) = 50$$

$$V = 10000\pi h$$

$$\frac{dV}{dt} = 10000\pi \frac{dh}{dt}$$

$$\frac{dh}{dt} = \frac{1}{10000\pi} \frac{dV}{dt}$$

$$\frac{dh}{dt} = \frac{-7}{10000} \frac{\text{cm}}{\text{s}}$$

5 Cones

Problem 7. Sand pours onto a surface at $15\text{cm}^3/\text{s}$, forming a conical pile with a base diameter that is always equal to the pile's altitude. How fast is the altitude of the pile increasing when the pile is 8cm high?



$$V = \frac{\pi}{3} r^2 h$$

$$V = \frac{\pi}{12} h^3$$

$$2r = h$$